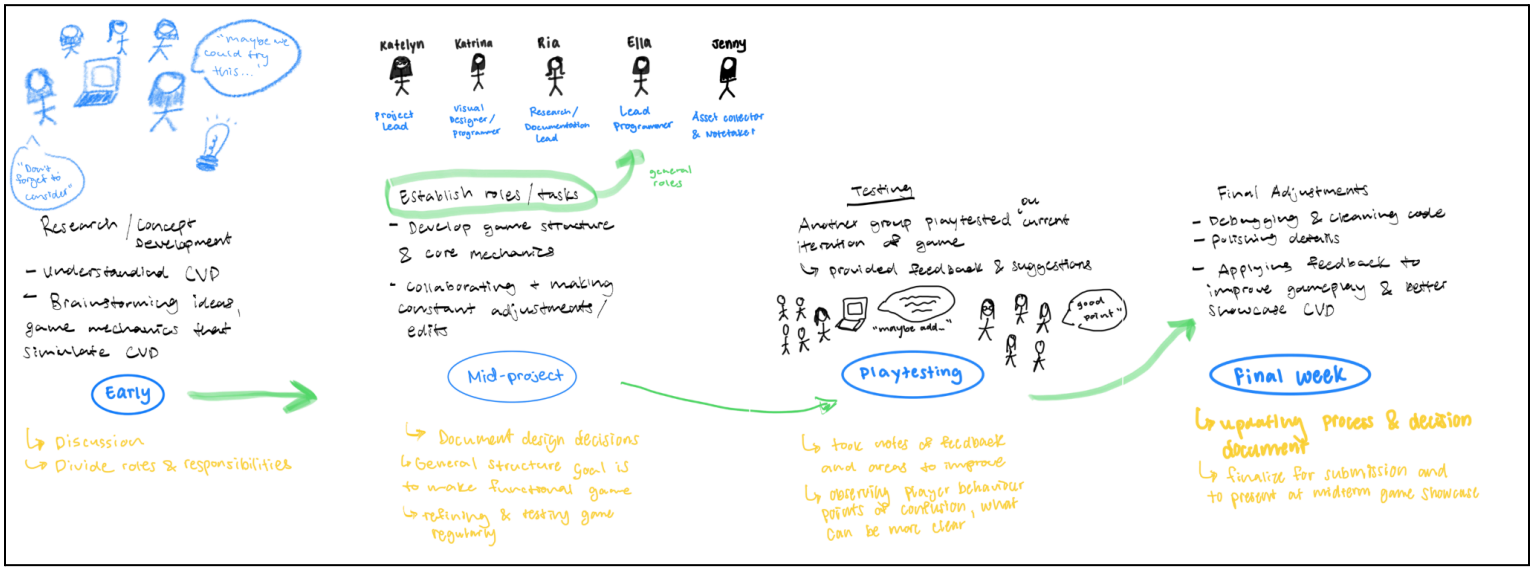
A3 Process & Decision Documentation

Process Overview Visualization



Project/Assignment Decisions

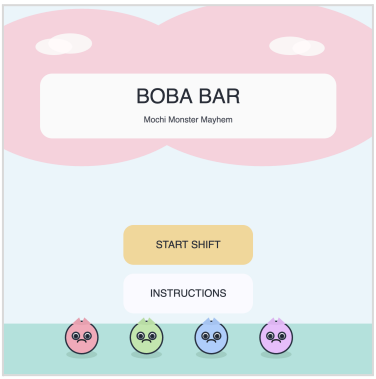
In Assignment 3, we developed our game prototype aimed at teaching non-colour blind individuals about colour vision deficiency and simulating what it would be like to possess it. GenAI was used to support the development of our game code, ensuring we maintained a consistent and smooth gameplay overall. Through our past playtesting feedback, we implemented the following improvements to our game:

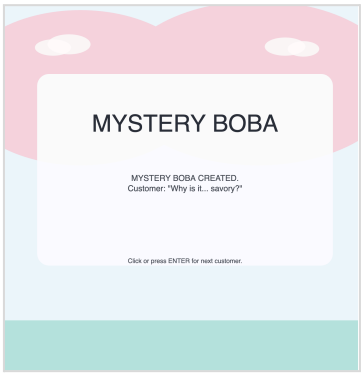
- Added in a tutorial level that introduced game mechanics and affordances in a simple manner
    - Making instructions more clear and understandable to reduce initial confusion
- Fixing the vision feature so it automatically switches to CVD mode when timer runs out, making colour deficiency more visible
- Level upgrades: as each round passes, progressively adding more features and options, new vision levels and mini activities relation to colour vision deficiency + varying ingredients per drink with not all orders having the same options
- More appealing layout and visuals with custom-made backgrounds for start and order screens
- Added audio and sound effects to elevate game experience and feedback

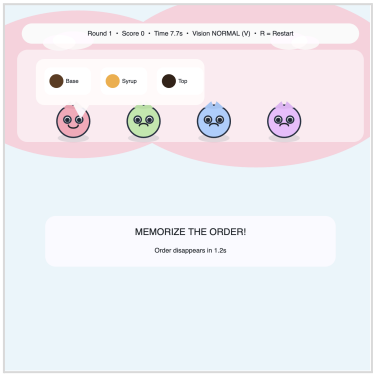
Overall, we focused on developing our game through a stronger storyline, enhanced levels, and stronger game mechanics in order to better showcase/teach players about colour vision deficiency and doing so in a way that makes the learning feel entertaining as well.

Role-Based Process Evidence

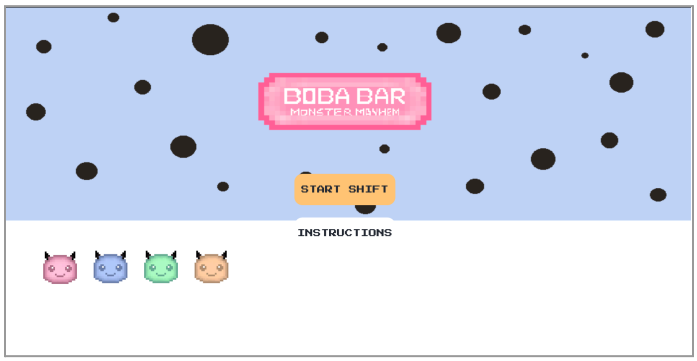
(first game)

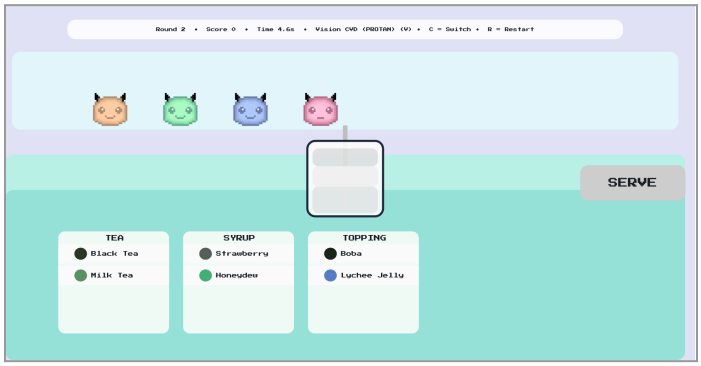


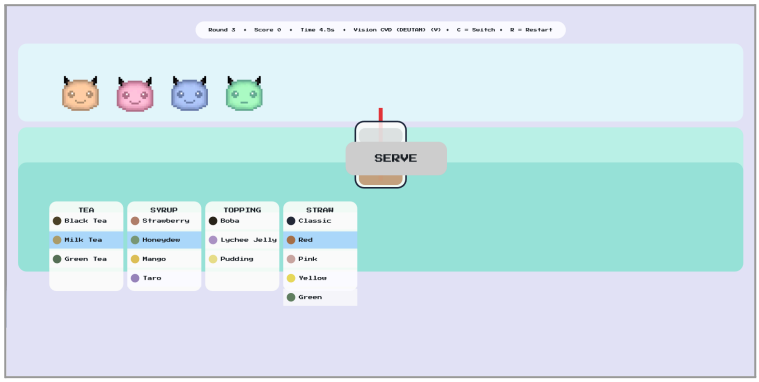


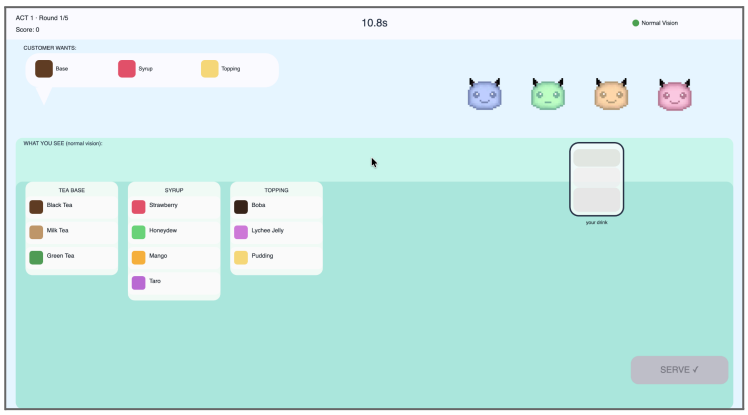


(work-in-progress)

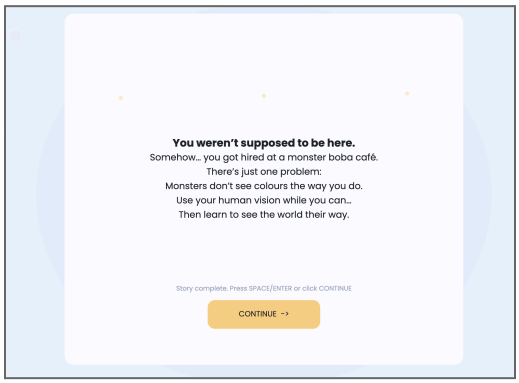


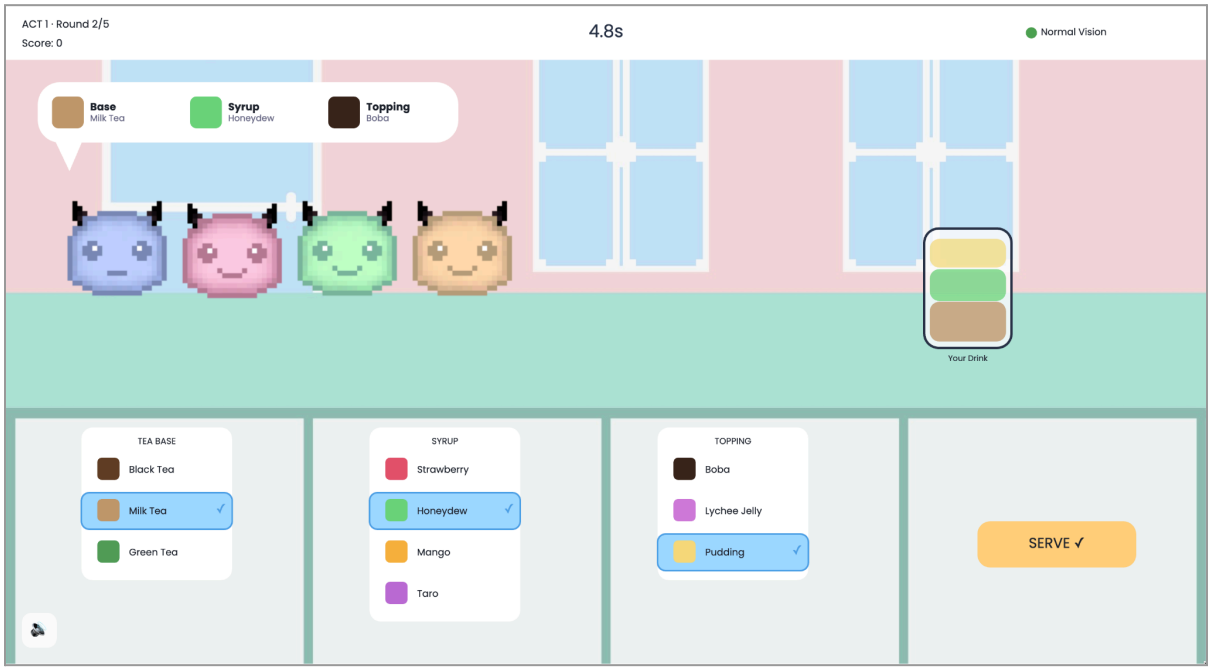


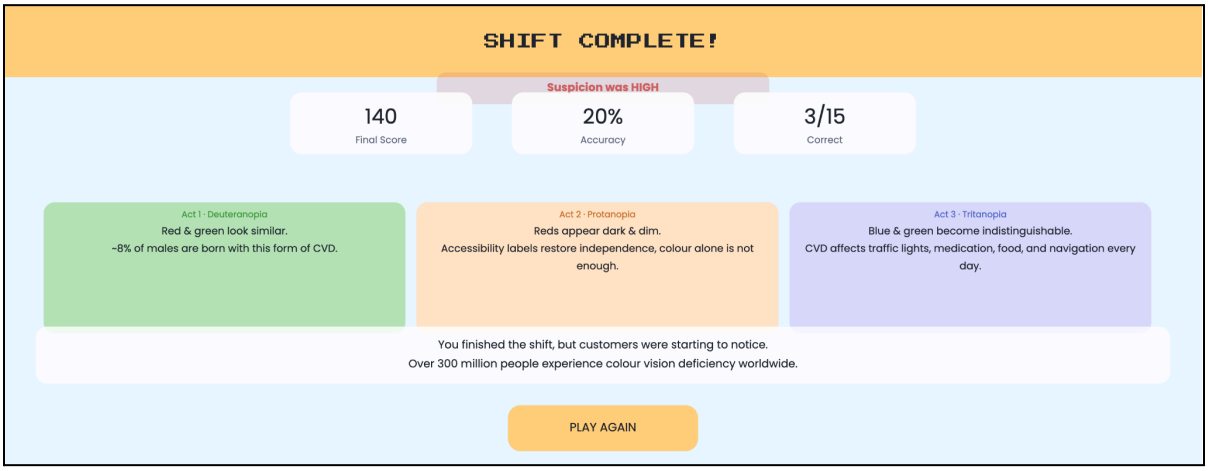




(final iteration)







Goal of Work Session

- Brainstorm game concept and mechanics related to colour vision deficiency
- Build and code game prototype to reflect proposed ideas
- Debug and refine code to ensure proper level progression
- Adjust visual elements and overall appearance
- Revise key mechanics based on playtesting feedback
- Fix areas or errors affecting gameplay experience
- Document suggestions and decisions for future reference

Tools, Resources, or Inputs Used

- ChatGPT 5.2
- Lecture notes and class examples
- Teammates / peer support
- Playtesting feedback

GenAI Documentation

Date Used: Apr 2, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: To brainstorm ideas relevant to our disability and help develop code for our final game prototype.

Summary of Interaction: We provided GenAI with our initial game state and asked to help interpret information from the National Eye Institute source on types of colour vision deficiency. This includes how different conditions affect colour perception like:

- Red-green deficiencies
- Blue-yellow deficiencies
- Complete colour deficiencies

GenAI translated the information into actual coding by giving us examples of simplified RGB adjustments and visual changes that could be implemented into our final gameplay. This allowed our game to be more accurate on how colours are being reflected in different conditions.

Human Decision Point(s): We made the final visual decisions of the game by editing and cleaning shape and text alignment, font type, and colours for a more cohesive and appealing look. While GenAI assisted in interpreting research, we determined how those findings would be applied in the game. This included selecting which types of colour vision deficiency to simulate,

adjusting the intensity of colour changes, and integrating these effects into gameplay progression.

We also made all final decisions regarding mechanics, visuals, audio, and overall user experience based on playtesting and feedback.

Integrity & Verification Note: We checked GenAI for accuracy, bias, and made sure it was appropriate to fit for this assignment. We made sure the output aligned with the class materials and verified it by rereading the assignment instructions and comparing them to GenAI suggestions. Any content that was beyond what we learned in class was removed and replaced.

Scope of GenAI Use: GenAI was mainly used during the ideation/starting stage and for outlining the basic structure of the game. The final design choices, testing, and adjustments to the mechanics were done by us.

Limitations or Misfires: Limitations when clicking or interacting with certain actions. Sometimes the tool could not code certain actions properly or as we intended/expected, and this is where we had to debug. Additionally, translating research-based colour models into working code required manual adjustment and testing to ensure visual accuracy within the game environment.

Summary of Process (Human + Tool)

We explored several concepts that teach colourblindness through mechanics (puzzles, matching, reaction games) and chose bubble tea assembly because it supports memory, time pressure, and colour-based selection.

Throughout the playtesting and midterm check-ins we understand now that the colour vision deficiency aspect was not represented properly in the initial iteration. As a result, we refined this approach by integrating research-based colour vision deficiencies and embedded them right into the gameplay. Using information from the National Eye Institute, we adjusted the RGB colours to better simulate different types of colour vision deficiencies and we expanded on adding more levels, ingredients, and audio progressions to improve engagement.

Overall, this process focused on the improvement of having accuracy within the disability representation and the overall gameplay experience.

Decision Points & Trade-offs

We considered how to best represent colour vision deficiency while keeping the game playable and engaging. Originally, we had a toggle system that allowed players to switch between normal vision and CVD mode. Based on feedback, we removed this and instead implemented an automatic transition during gameplay to make the effect more noticeable and integrated. Another decision involved how accurately to simulate colour vision deficiency. While research from the National Eye Institute guided how colours should be transformed, fully accurate simulations made it difficult for players to distinguish between options. We adjusted the intensity of the colour changes to balance realism with usability.

We also decided to expand the mechanics by adding more ingredients, level progression, and audio. While this increased engagement, it required careful adjustments to ensure the game did not become too complex or overwhelming.

Verification & Judgement

To verify our decisions and justify our changes, we all collectively playtested the game together. As each member took turns playing the game, we were able to find small bugs and visual aspects to be changed and enhance the gameplay. We also compared our implementation to the National Eye Institute source to ensure that the types of colour vision deficiency we represented were accurate. While the simulation was simplified, we made sure it still reflected key differences in colour perception.

Notably, we found that some of the shapes and text were not aligning together and that the font was too basic, so we changed these factors and it made the overall game more smooth and visually appealing. Through testing, we found that the updated version improved gameplay flow, difficulty progression, and overall clarity, making the experience more engaging and aligned with the assignment goals.

Limitations, Dead Ends, or Open Questions

Although our game improved, there are still areas that can be refined. One limitation was technical inconsistencies across different screens, where UI elements like the serve button would shift depending on the monitor. This required manual adjustments to maintain alignment. Many elements also had to be positioned manually, which was time intensive and made updates less efficient. We also encountered caching issues in Live Server, where changes did not always reflect immediately, requiring extra steps to verify updates.

From a design perspective, the colour transformations are still simplified and balancing difficulty remains a challenge, as some colours are either too easy or too difficult to distinguish.

Appendix

Initial RGB adjustment

Prompt:

can u make it so that the colours follow these rules and goes monocrome at the same time https://www.nei.nih.gov/eye-health-information/eye-conditions-and-diseases/color-blindness/types-color-vision-deficiency

Output:

Yes — the clean way is:

1. **apply an NEI-inspired CVD shift first**
2. **then fade that result toward grayscale each round**

That matches the NEI descriptions pretty well:
deuteranomaly makes some greens look redder, protanomaly makes some reds look greener and less bright, tritan defects affect blue/green and yellow/red distinctions, and full monochromacy means no color at all. This code is still an approximation for gameplay, not a medical simulator.

Replace your current getShownColor() section in game.js with this:

```
// choose which deficiency you want to simulate
let cvdType = "DEUTAN"; // "DEUTAN", "PROTAN", or "TRITAN"

function getMonochromeFactor() {
  let factor = (round - 1) * 0.1;

  if (visionMode === "CVD") {
    factor += 0.15;
  }

  return constrain(factor, 0, 0.92);
}

function applyCVD(rgb) {
  let r = rgb[0];
  let g = rgb[1];
  let b = rgb[2];

  if (visionMode !== "CVD") {
    return [r, g, b];
  }

  if (cvdType === "DEUTAN") {
```

```
    // greens shift toward red
    let rg = (r + g) / 2;
    g = lerp(g, rg, 0.75);
    r = lerp(r, rg, 0.35);
  }

  else if (cvdType === "PROTAN") {
    // reds shift toward green and look less bright
    let rg = (r + g) / 2;
    r = lerp(r, rg, 0.75);
    g = lerp(g, rg, 0.35);
    r *= 0.82;
  }

  else if (cvdType === "TRITAN") {
    // blue-green confusion, plus some yellow-red confusion
    let bg = (b + g) / 2;
    b = lerp(b, bg, 0.75);
    g = lerp(g, bg, 0.35);

    let yr = (r + g) / 2;
    r = lerp(r, yr, 0.18);
  }

  return [
    constrain(r, 0, 255),
    constrain(g, 0, 255),
    constrain(b, 0, 255),
  ];
}

function getShownColor(rgb) {
  let cvd = applyCVD(rgb);

  // grayscale fade after CVD shift
  let gray = 0.299 * cvd[0] + 0.587 * cvd[1] + 0.114 * cvd[2];
  let mono = getMonochromeFactor();

  return [
    lerp(cvd[0], gray, mono),
    lerp(cvd[1], gray, mono),
    lerp(cvd[2], gray, mono),
  ];
}
```

Recheck RGB adjustment

Prompt:

This is what each mode looks like. Do these align correctly with what each type of colour vision deficiency actually sees?

Output:
Short answer: **you're on the right track, but these are NOT fully accurate to real CVD from the NIH site**

What's NOT accurate (compared to NIH)

According to the NIH reference:

1. PROTAN (red deficiency)

Real effect:

- Reds appear **darker / almost black**
- Red–green become hard to distinguish

Your version:

```
r = lerp(r, rg, 0.75);
r *= 0.82;
```

Issue:

- Not dark enough
- Still too colourful

Fix (better):

```
r *= 0.6; // stronger darkening
```

2. DEUTAN (green deficiency)

Real effect:

- Red and green look very similar
- BUT brightness stays mostly normal

Your version:

```
g = lerp(g, rg, 0.75);
r = lerp(r, rg, 0.35);
```

This is actually **pretty good**
Just slightly too subtle

3. TRITAN (blue deficiency)

Real effect:

- Blue ↔ green confusion
- Yellow ↔ red confusion
- Very rare condition

Your version:

```
let bg = (b + g) / 2;
b = lerp(b, bg, 0.75);
g = lerp(g, bg, 0.35);
```

Issue:

- Needs **stronger shift**
- Right now still too distinguishable